

Aditya Santosh Siras

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ABOUT

Full Stack & AI/ML Engineer (CS grad, June 2026) with production-scale experience shipping systems handling real institutional and organizational traffic. Published researcher at IEEE InGARSS 2025. Expertise: Building scalable web platforms (React/Node.js/TypeScript), LLM/RAG systems (LangChain, vector DBs, fine-tuned NER), and secure payment infrastructure. Track record: Bhuvan GeoIntel (ISRO, peer-reviewed), LOCO-INFO (Ministry of Railways), CREA platform (500+ active users, live), SSGMCE institutional website (150+ pages, live). Passionate about end-to-end ownership and shipping production-grade systems.

EXPERIENCE

AI/ML Engineer Intern

Feb. 2025 – April 2025

National Remote Sensing Centre (ISRO), Shadnagar, Telangana

- **Developed Bhuvan GeoIntel**: Geospatial intelligence extraction system for PDFs. Published at **IEEE InGARSS 2025** (peer-reviewed, international conference).
- Fine-tuned **spaCy NER model** on Indian location names using rule-based techniques + **Geopy** for accurate geographic mapping. Achieved high precision on noisy PDF data.
- Built **RAG (Retrieval-Augmented Generation) system**: LangChain + Mistral-7B-Instruct + nomic-embed-text embeddings + ChromaDB for context-aware query responses. Integrated Google PSE + web scraping.
- Mapped geolocations on **ISRO Bhuvan WMS layers** using **OpenLayers**, enabling interactive spatial analysis for decision-makers on national-level geospatial planning.
- Tech: Python, LangChain, spaCy, ChromaDB, Mistral-7B, OpenLayers, Geopy. Demonstrated ability to handle complex real-world data with production-grade ML pipelines.

App Developer Intern

July 2025 – Aug. 2025

Electric Loco Shed, Bhusawal – Central Railway (Ministry of Railways)

- Developed **LOCO-INFO**: Railway shed management system for locomotive maintenance scheduling. Implemented in **Flutter** with **Firebase Firestore** real-time database.
- Built real-time synchronization system for live locomotive data across platforms using **Firebase Firestore (Realtime DB)**. Ensured data consistency across distributed endpoints.
- Implemented **secure authentication** and **role-based access control** to ensure data integrity, user accountability, and organizational trust in production railway operations.
- Digitized manual record-keeping, enabling faster maintenance scheduling and improved operational efficiency for the railway shed.

Full Stack Developer

Aug. 2024 – June 2026

Central Railway Engineers Association (CREA), Bhusawal

- Designed and shipped MERN platform to 500+ active users (live at **crea.org.in**). Complete end-to-end ownership: database schema, REST API, React frontend, deployment pipeline.
- Engineered **Razorpay payment integration** for membership fees + renewals with **idempotent webhook handling** to prevent double-charging on network retries. Secure PCI-DSS compliant flow.
- Built admin CMS with **role-based access control (RBAC)**, JWT authentication, visual page editor, analytics dashboard, dynamic management of news/events/faculty/notices.
- Implemented multi-step form logic, automated email notifications, PDF certificate generation, batch user uploads. Reduced administrative processing time by 78%.
- Tech: React, Node.js/Express.js, MongoDB, JWT Auth, Razorpay API, CI/CD pipelines.

Full Stack Developer

Jan. 2026 – May 2026

Shri Sant Gajanan Maharaj College of Engineering (SSGMCE), Shegaon

- Designed and deployed **150+ page institutional portal** for SSGMCE (live at **ssgmce.ac.in**). Production-grade platform serving institutional traffic with **98% uptime**.
- Tech stack: **React 18, TypeScript, Tailwind CSS, Node.js, MongoDB, JWT Authentication, CI/CD pipelines**. Optimized for performance and accessibility.
- Built proprietary Admin CMS with **role-based access control**, visual page editor, analytics dashboard, and dynamic content management for news, events, faculty, notices, documents.
- Implemented institutional compliance sections: IQAC, NIRF, NBA, NAAC with structured data and reporting. Multi-departmental infrastructure supporting all academic systems, timetables, syllabi, UG/PG/PhD admissions.
- Features: Facilities section (hostels, library, sports, student orgs), student life portal, secure backend with REST API architecture, role-based access control (RBAC), secure file upload & serving.

PERSONAL PROJECTS

- MemoryOS** | *Memory-as-a-Service API for LLMs (In Development)* Personal Project
- Building a production-scale memory layer for LLMs. **FastAPI backend** with **Pinecone vector DB** for per-user memory isolation (multi-tenant architecture).
 - **Next.js dashboard** for memory management + analytics. **Razorpay-gated API key generation** for monetization. **MCP server integration** for Claude Desktop.
 - Demonstrates: Production system design, multi-tenancy patterns, payment integration, vector DB semantics, secure API architecture.

EDUCATION

Shri Sant Gajanan Maharaj College of Engineering (SSGMCE) Shegaon, Maharashtra
Bachelor of Engineering, Computer Science Engineering — CGPA: 8.35/10.0 June 2026

PROJECTS

- Bhuvan GeoIntel** | *Python, spaCy, LangChain, ChromaDB, OpenLayers* IEEE InGARSS 2025 (Published)
- Geospatial intelligence extraction system for PDFs using fine-tuned spaCy NER, RAG-based retrieval with LangChain, and interactive visualization on ISRO Bhuvan WMS maps. Peer-reviewed publication.
- Loco-Info** | *Flutter, Firebase Firestore, Real-time DB, Authentication* Ministry of Railways
- Railway shed management system with real-time locomotive data synchronization, secure authentication, role-based access control, and live data consistency across platforms.
- AI-ML Based Career Counseling System** | *Python, ML, Google Firebase, Dialogflow* Smart India Hackathon 2023 (Grand Finale)
- AI-ML platform guiding secondary students in career choices through personalized assessments. Built with Python, HTML/CSS/JS, Google Firebase, and Dialogflow AI API.
- Personal Finance Tracker App** | *Flutter, Firebase*
- Cross-platform finance tracking app with user authentication, expense tracking, data persistence using Firebase. Clean architecture with state management.

TECHNICAL SKILLS

Frontend: React 18, TypeScript, Next.js, Tailwind CSS, HTML/CSS, JavaScript, Flutter
Backend: Node.js, Express.js, FastAPI (Python), REST APIs, JWT Authentication, Secure APIs
Databases & Storage: MongoDB, PostgreSQL, Firebase Firestore, Pinecone (Vector DB), ChromaDB
AI/ML & LLMs: LangChain, Mistral-7B, spaCy NER, ChromaDB, Embeddings, RAG Systems, Fine-tuning
Mobile: Flutter, Dart, Firebase, Kotlin, Android, Jetpack Compose, MVVM
DevOps & Tools: Git, Docker, CI/CD Pipelines, Google Cloud Platform, Google Firebase, VS Code
Data Science: pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, Python

PUBLICATIONS & ACHIEVEMENTS

Peer-Reviewed Publication: "An Integrated Framework for Geospatial Content Extraction and Semantic Summarization from PDF Documents" Accepted at **IEEE InGARSS 2025**, IIIT Bhubaneswar (Dec. 2025)

Shortlisted in **Smart India Hackathon Grand Finale 2023** for "AI-ML Based Career Counseling System"

Internal Hackathon Winner, Smart India Hackathon 2024

Exceptional Performance Award, Code Vipassana Season 5 (Duet AI)

CERTIFICATIONS

Machine Learning Specialization: Stanford University & DeepLearning.AI (Coursera)
Deep Learning Specialization: Stanford University & DeepLearning.AI (Coursera)
Python for Data Science: NPTEL
Exceptional Performance in Code Vipassana Season 5: Duet AI